

Q: Compare Combinational and Sequential circuits.

Sr No	Parameter	combinational circuit.	sequential circuit
(1)	Output Depend on	input present at that instant of time.	Present input and past inputs/outputs
(2)	Memory	Not Required	Required.
(3)	clock input	Not Required	Required
(4)	Example	Adder, subtractor, code converter	Flip Flop, shift Register counter.
(5)	Speed	simple Fast	slow
(6)	Design	Simple	complex

Q: What is triggering. Explain types of Triggering.

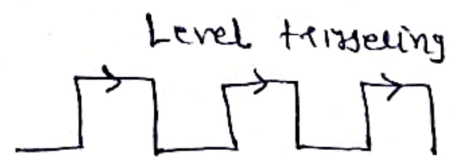
→ in sequential ckt output will response only when applying input clock pulse. Applying clock pulse is known as triggering.

→ There are two types of triggering.

- (1) Level triggering
- (2) Edge triggering.

* Level triggering:-

in Level triggering O/P changes during the high voltage period or low voltage period.

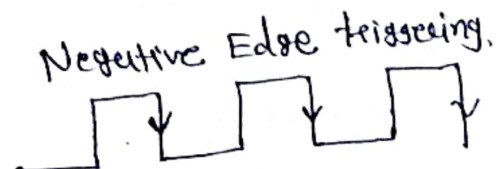
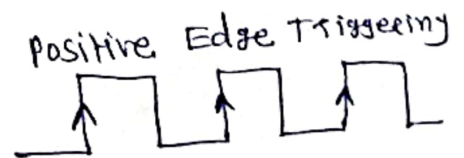


There are two types of level Triggering.

- (1) Positive level triggering
- (2) Negative level triggering.

* Edge triggering:-

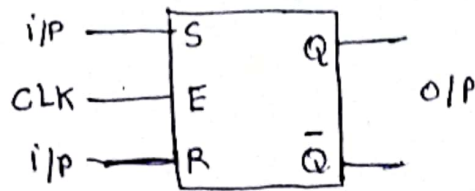
O/P changes during transistion low to high or high to low



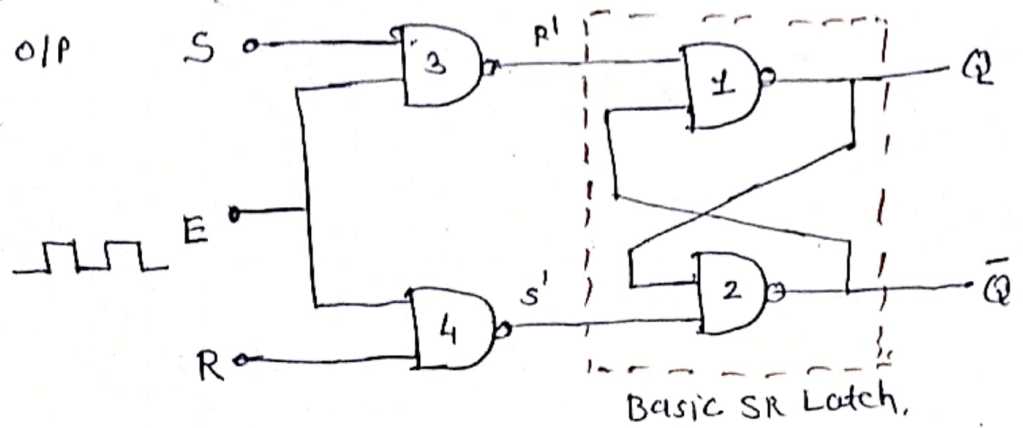
There are two types of Edge triggering.

- (1) positive Edge triggering.
- (2) Negative Edge triggering

Q1- Level Triggered S-R Flipflop



Symbol



- Clock SR Flipflop consists of Basic SR Latch and two other NAND gates to provide clock pulse as shown in fig.
- Output will respond to the input when clock pulse is enable.

→ Truth table

Input				Output		Comments
case	Enable	S	R	Q_{n+1}	$\bar{Q}_{n+1}$	
1	0	X	X	Q_n	$\bar{Q}_n$	No change
2	1	0	0	Q_n	$\bar{Q}_n$	No change
3	1	0	1	0	1	Reset
4	1	1	0	1	0	Set
5	1	1	1	Indeter minite	indeter minite	

Operation

Case 1:- $E=0, S=X, R=X$

Since $E=0$, Gate 3, 4 output = 1
so, $s', r'=1$, Q will not change.

Case 2:- $E=1, S=0, R=0$

If $S=R=0$, Gate 3, 4 output = 1
 $s', r'=1$, Q will not change.

Case 3:- $E=1, S=0, R=1$

$S=0$, output of NAND 3 $r'=1$
as $R=1$ output of NAND 4 $s'=0$

$Q_{n+1} = 0$, This is Reset condition.

Case 4:- $E=1, S=1, R=0$

output of NAND 3 $r'=0$
output of NAND 4 $s'=1$

$Q_{n+1} = 1$. This is set condition.

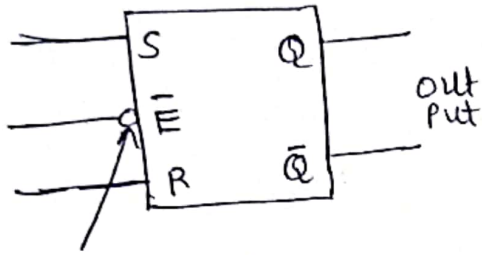
Case 5:- $E=1, S=1, R=1$

output of NAND 3 & 4 both = 0
 $s', r'=0$

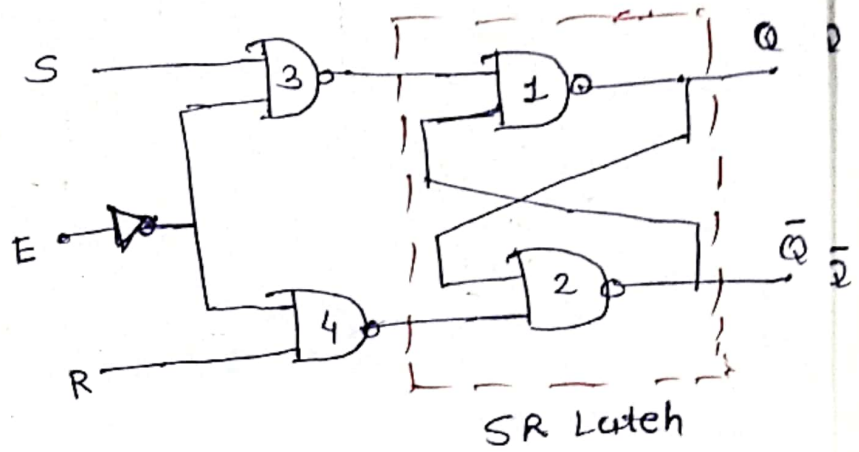
This is invalid output

'Race' condition occurs

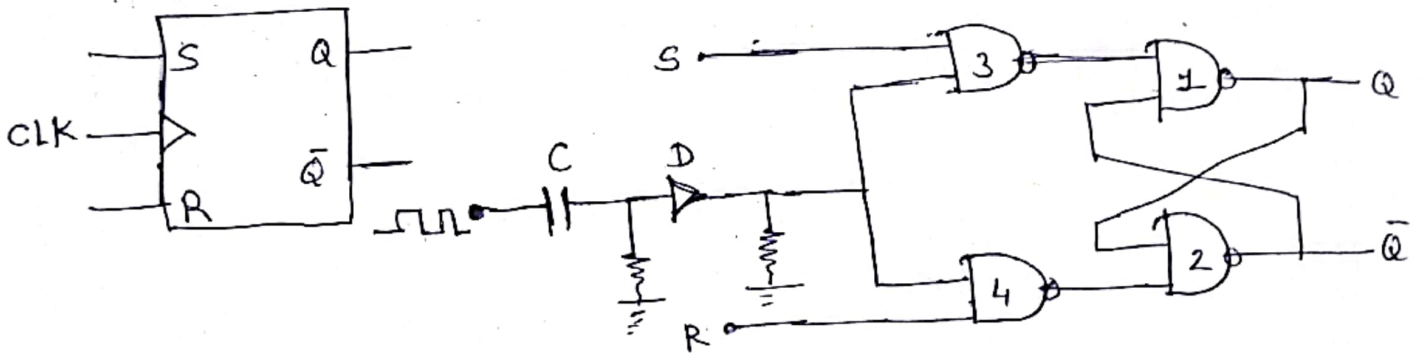
Negative level trigger SR Flip Flop



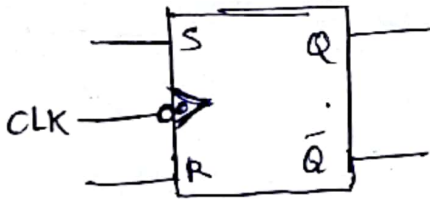
Bubble represent
Low level triggering



* Positive Edge triggered SR Flip Flop



* Negative Edge triggered SR Flip Flop.



Diode direction will reverse ~~from~~ ⁱⁿ
above ckt
For Negative Edge trigger.

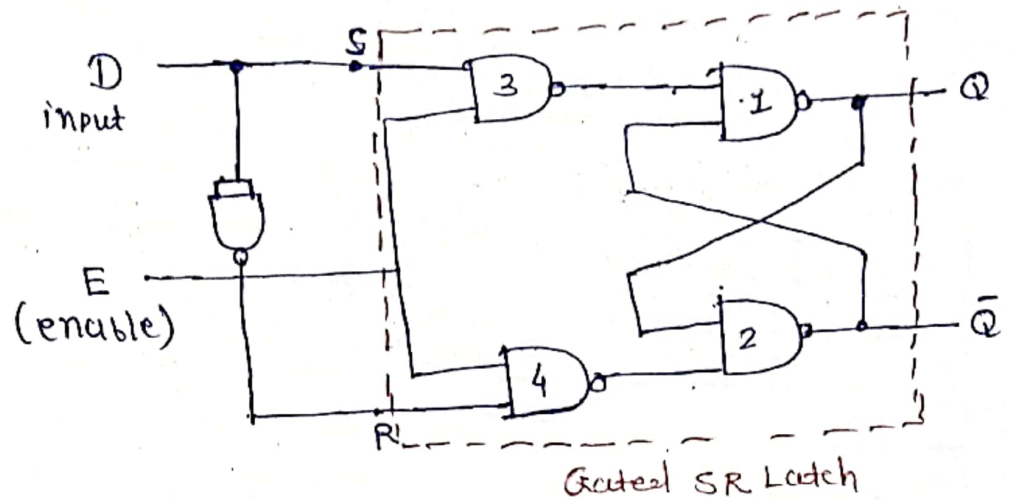
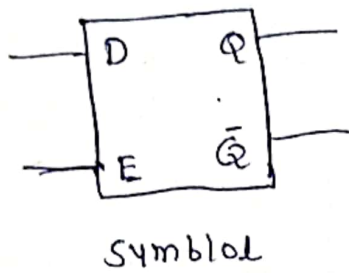
Q:- what is Disadvantage of SR Flip flop.

Ans:- when $S = R = 0$ the o/p don't change

&
when $S = R = 1$, output is invalid due to race condition

This disadvantage is overcome in D flip flop.

Level Triggered D Flip Flop



Level Triggered D Flip Flop shown in fig.

- it is simple Level triggered SR Flip Flop with NAND inverters between S & R. so the S & R will always complement of each other.
- this will avoid problem of $SR=00$ and $SR=11$ condition.

Truth table

input		output		Comments
E	D	Q_{n+1}	$\bar{Q}_{n+1}$	
0	X	Q_n	$\bar{Q}_n$	No change
1	0	0	1	Reset
1	1	1	0	set

Operation

Case 1:- $E=0$, For any value of D, No change in o/p.

Case 2:- $E=1$, $D=0$ Then $S=0$ and $R=1$
Then $Q_{n+1}=0$ and $\bar{Q}_{n+1}=1$ This is Reset condition.

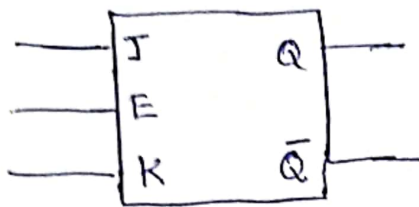
Case 3:- $E=1$, $D=1$ Then $S=1$, and $R=0$

Then $Q_{n+1}=1$ and $\bar{Q}_{n+1}=0$, This is Set condition.

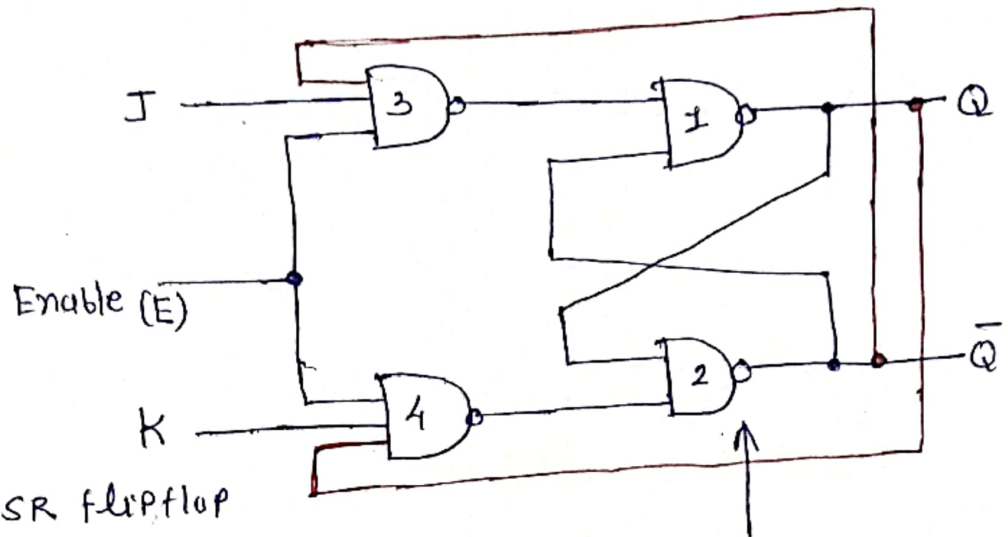
From Truth table, it shows That Q output will follow D input

$$\begin{aligned} D=0 &\Rightarrow Q=0 \\ D=1 &\Rightarrow Q=1 \end{aligned}$$

Level Triggered JK Flipflop



Symbol



Basic SR latch

→ it consists Basic SR flipflop and an enable input with Q and $\bar{Q}$ are returned back and connected to the input of NAND 4 and 3 respectively.

JK FLIPFLOP

Truth table

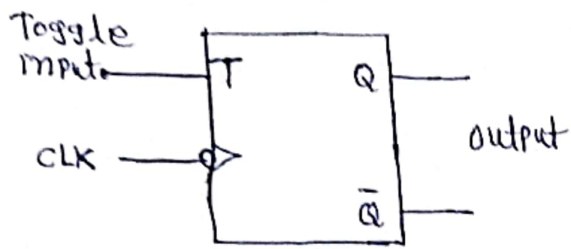
input			output		Comments
E	J	K	Q_{n+1}	$\bar{Q}_{n+1}$	
0	X	X	Q_n	$\bar{Q}_n$	No change
1	0	0	Q_n	$\bar{Q}_n$	No change
1	0	1	0	1	Reset
1	1	0	1	0	set
1	1	1	$\bar{Q}_n$	Q_n	Toggle

Operation:-

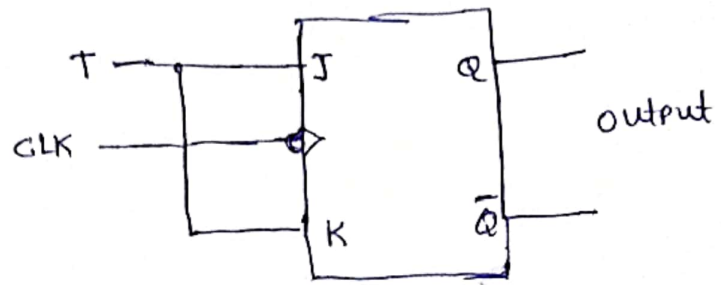
Case 1:-

Case 2:-

Toggle Flip Flop



Logic symbol



T Flip Flop

- In JK Flip Flop, J and K permanently connected together then it becomes T flip Flop
- it has one input T.
- Negative edge triggered T flip flop is shown in fig.

Truth table

input		output		comment
CLK	T	Q_{n+1}	$\bar{Q}_{n+1}$	
0	X	Q_n	$\bar{Q}_n$	No change
↑	X	Q_n	$\bar{Q}_n$	No change
↓	0	Q_n	$\bar{Q}_n$	No change
↓	1	$\bar{Q}_n$	Q_n	Toggle

Operation :-

Case 1:- when CLK = 0, 1 & ^{positive} ~~positive~~ edge & T = any value (X)
output won't change.

Case 2:- when CLK = Negative edge & T = 0
output won't change

Case 3:- when CLK = Negative edge & T = 1
output will toggle

Q:- what is Race around condition, write measures of preventing it.

or.

Q:- what is Race around condition, how to avoid it.

or

Q:- Explain Race around condition with wave form
(રસ અરાઉન્ડ કન્ડીશન અંગ્રેજી ? તો અડકાવાળા ઉપાય તમારો).

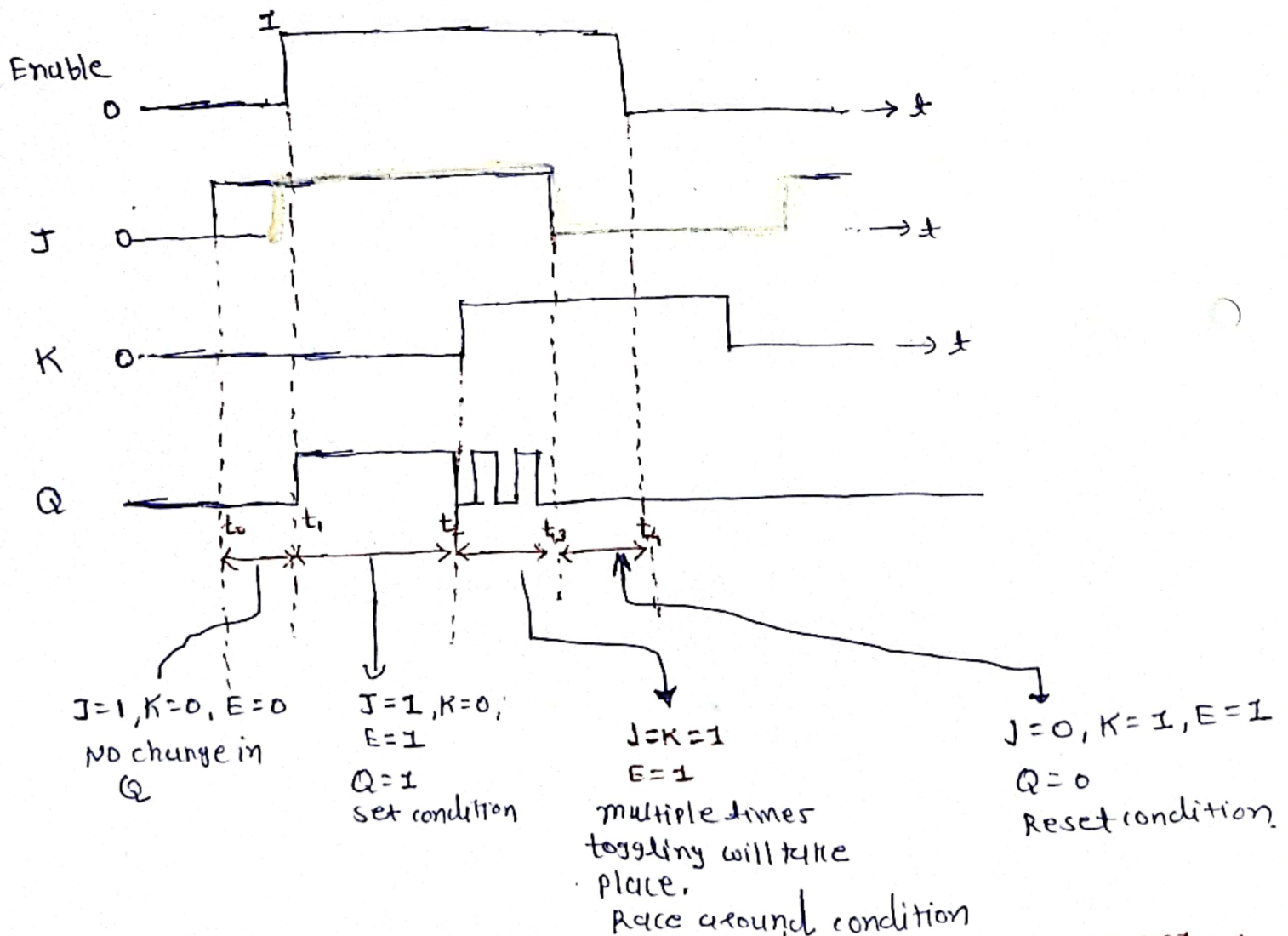
→ in Level triggered JK Flip Flop when $J=K=1$ then output Q will be in toggle mode. This is called Race around condition.

→ It can be avoided by

(1) using edge Triggered JK Flip Flop

(2) Master slave JK Flip Flop.

→ Wave Form

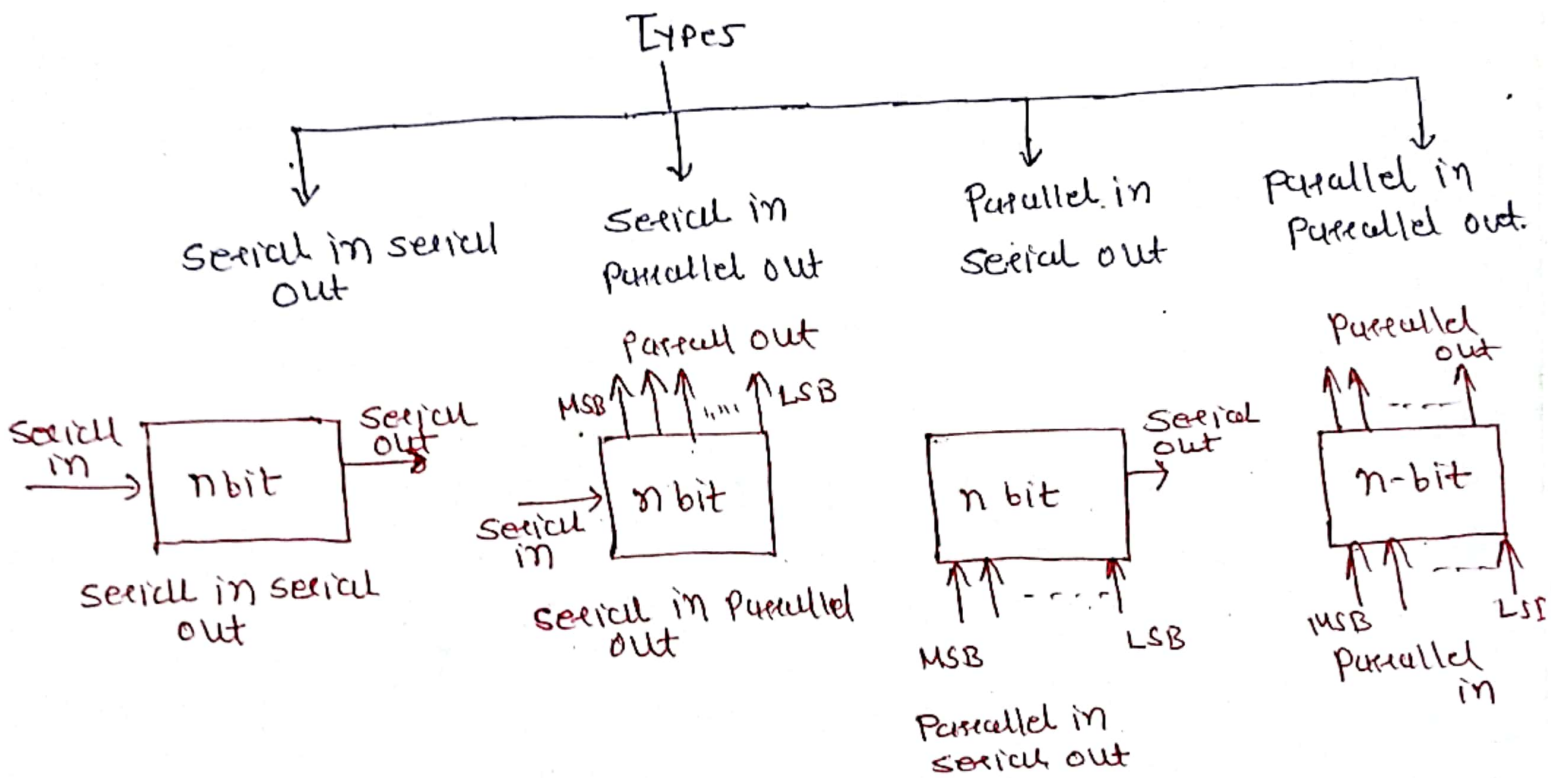


* Explain Binary to Gray code converter circuit.



Q what is Shift Register. Give types of shift Register,
Write Application of shift Register.

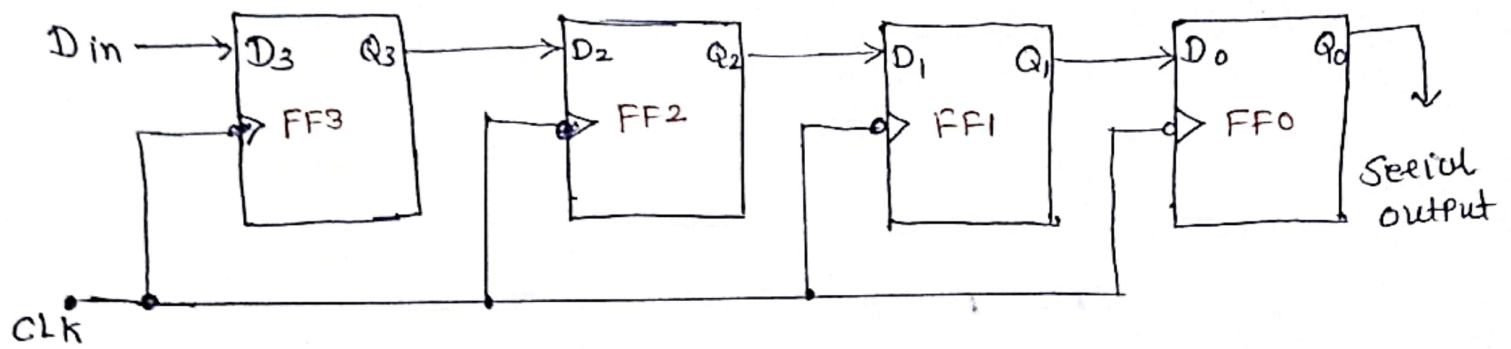
Shift Register - One flip flop can store 1 bit binary data.
To store n bit data, n flip flop required. Such a group of n flip flop is called Register.
When this stored data are shifted it is called shift Register.



Application of Shift Register

- (1) Temporary data storage
- (2) Delay line
- (3) Serial to Parallel converter
- (4) Parallel to serial converter
- (5) Ring counter
- (6) UP/Down counter
- (7) Data Transfer.
- (8) Multiplication & Division

Serial in Serial out shift Register (Shift Right Mode)



→ Din is given to FF3, output of First FF is given to the input of Next Flip Flop.

→ Clock pulse is given simultaneously to all Flip Flop.

Operation:- initially $Q_3 Q_2 Q_1 Q_0 = 0000$ and we want to shift 1111.

at First falling edge of clock, $Q_3 Q_2 Q_1 Q_0 = 1000$

apply next bit to Din. So $D_{in} = 1$

as next falling edge of clock, FF2 will set and

$$Q_3 Q_2 Q_1 Q_0 = 1100$$

Next bit to be stored $D_{in} = 1$

As next falling edge of clock FF1 will set and

$$Q_3 Q_2 Q_1 Q_0 = 1110$$

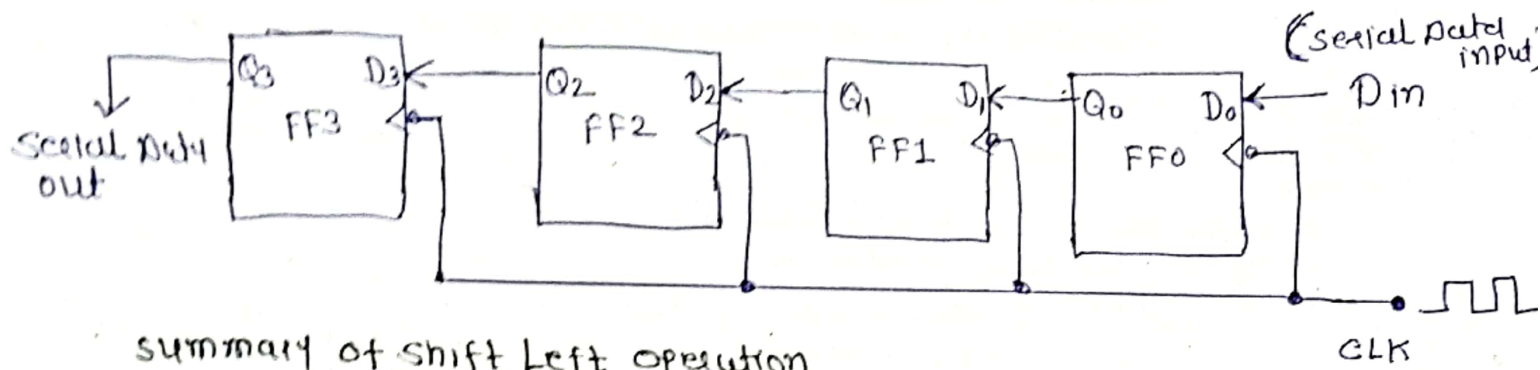
After fourth negative edge of clock, FF0 will set

$$Q_3 Q_2 Q_1 Q_0 = 1111$$

Summary of Shift Right operation

	CLK	D_{in}	Q_3	Q_2	Q_1	Q_0
initially			0	0	0	0
1 st	↓	1	1	0	0	0
2 nd	↓	1	1	1	0	0
3 rd	↓	1	1	1	1	0
4 th	↓	1	1	1	1	1

Q1: Serial In serial out Register (Shift Left mode)



summary of Shift Left operation

	CLK	Q ₃	Q ₂	Q ₁	Q ₀	D _{in}
Initially		0	0	0	0	
1 st	↓	0	0	0	1	1
2 nd	↓	0	0	1	1	1
3 rd	↓	0	1	1	1	1
4 th	↓	1	1	1	1	1

- D input is given to FF0. Output of first flip flop is given to the input of next flip flop.
- Clock pulse is given simultaneously to all flip flop.

operation :-

Initially $Q_3Q_2Q_1Q_0 = 0000$ and we want to shift 1111

at first falling edge of clock $Q_3Q_2Q_1Q_0 = 0001$ [FF0 = set]

apply next bit to Din so $D_{in} = 1$

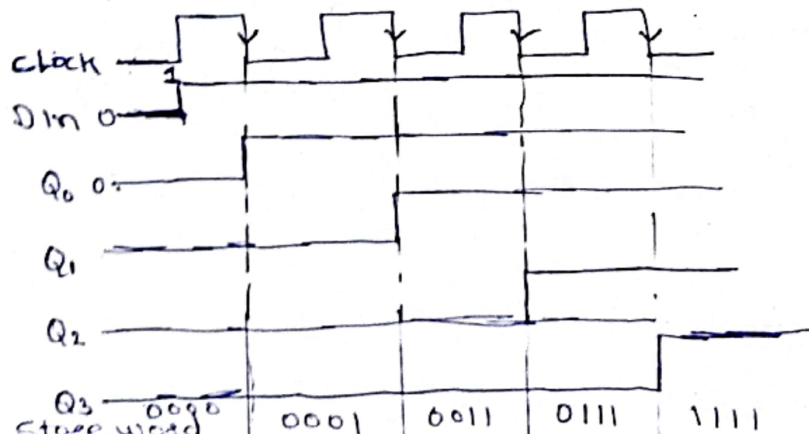
at Next falling edge of clock $Q_3Q_2Q_1Q_0 = 0011$ [FF0, FF1 = set]

Next Bit to be stored is $D_{in} = 1$

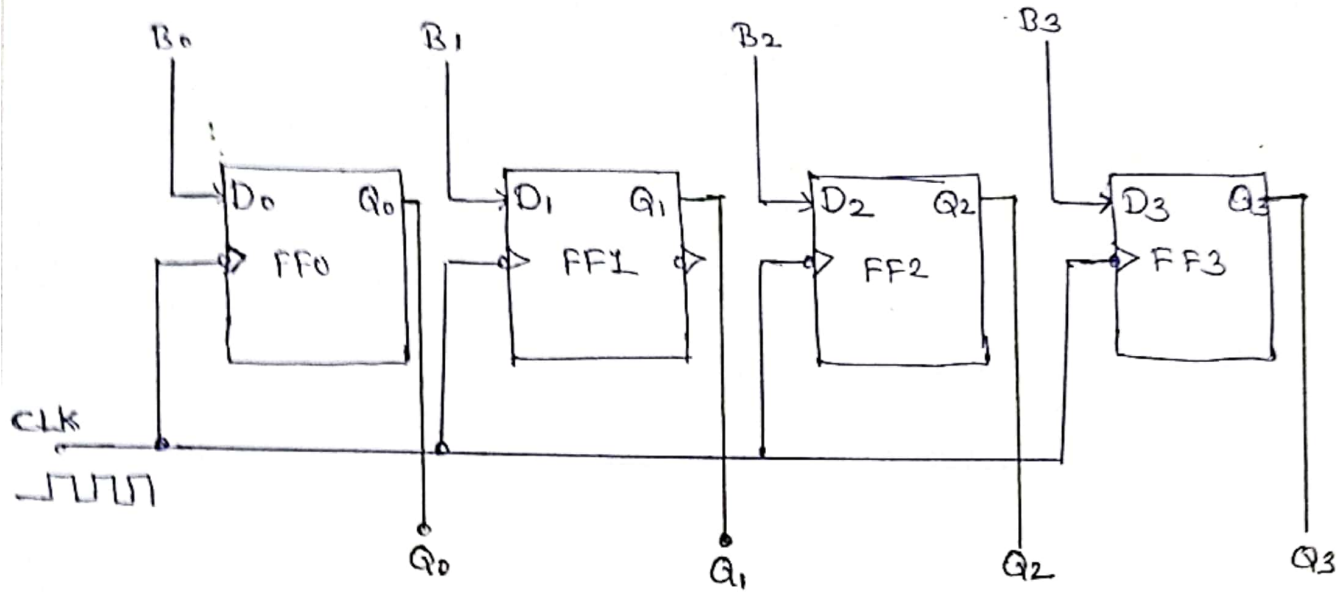
at Next falling edge of clock $Q_3Q_2Q_1Q_0 = 0111$ [FF0, FF1, FF2 = set]

After Fourth Negative edge of clock $Q_3Q_2Q_1Q_0 = 1111$ [All FF set]

Waveform

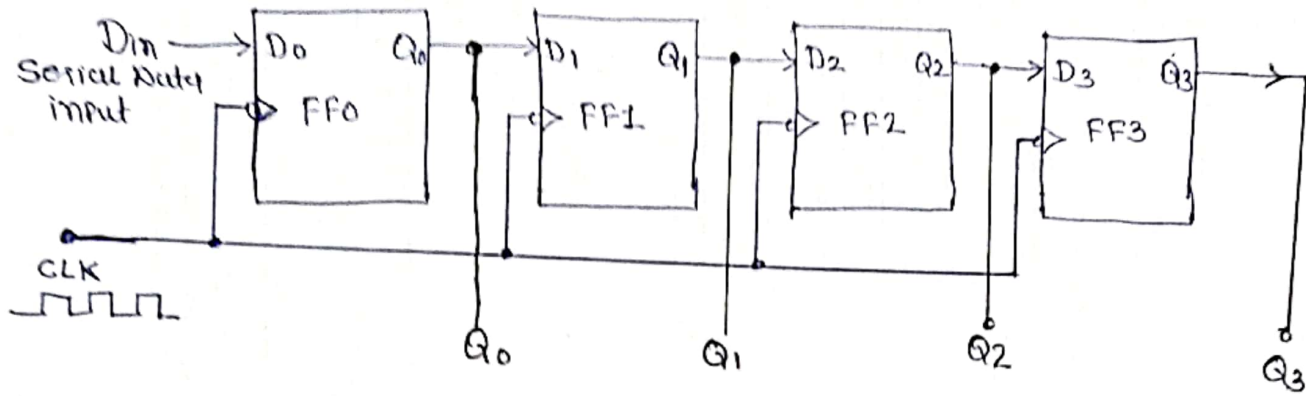


Parallel input parallel output Shift Register.



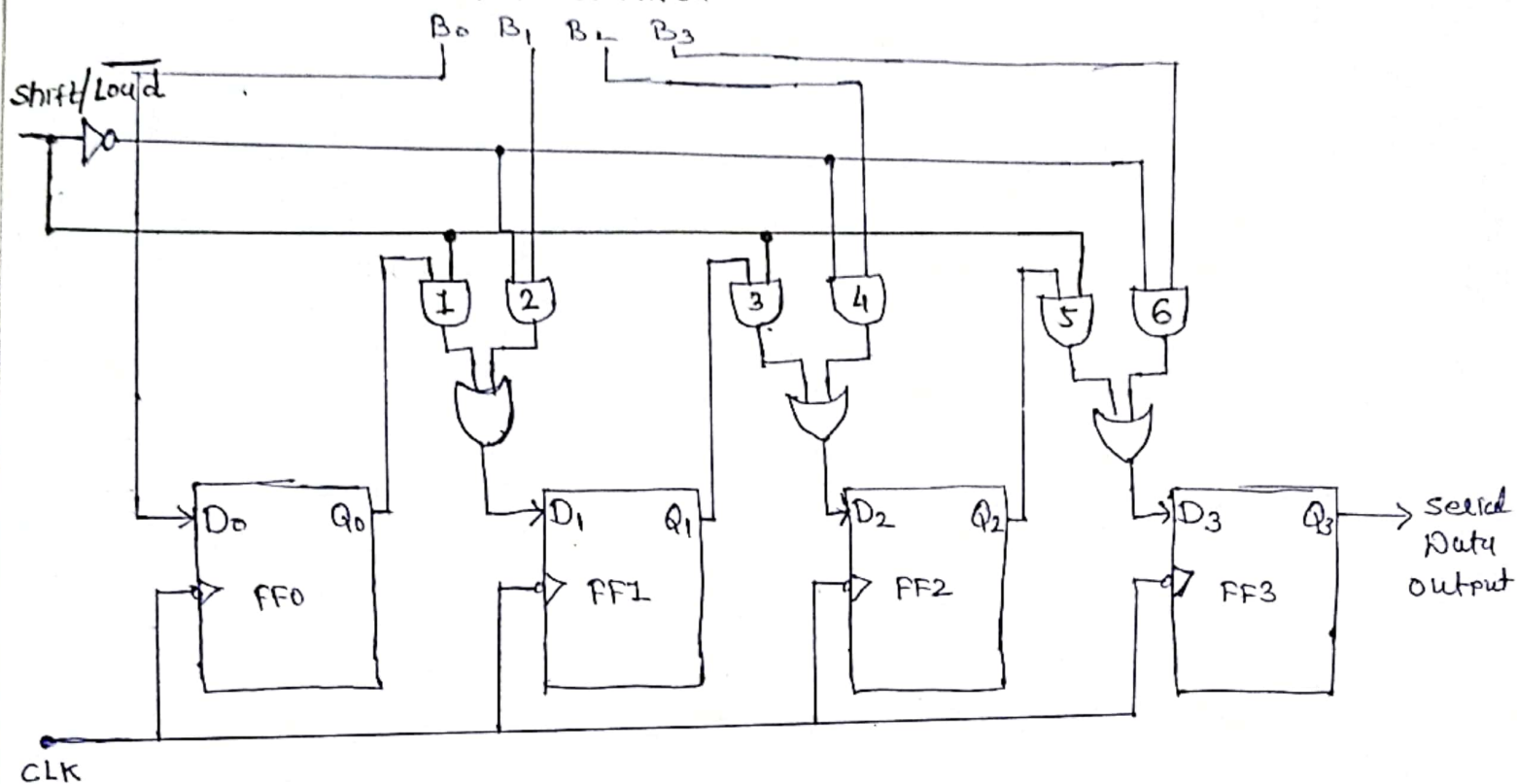
- Four Bit Binary B₀ B₁ B₂ B₃ is applied to the data input D₀ D₁ D₂ D₃ respectively of four flip flop.
- As soon as the negative clock edge is applied, the input Binary bits will be loaded in to the flip flop simultaneously.
- The loaded bits will appear simultaneously to the output side.
- Only one clock pulse is require to Load all the bits.

Serial in parallel out Register



- Data is entered serially and taken out parallel.
- First the data is loaded bit by bit, output are disabled as long as the loading is taking place.
- When loading is complete, the outputs are enabled so that all the loaded data is available simultaneously.
- To load 4 bit data, 4 cycles required. so speed is same as serial input serial output register.

Parallel input Serial out shift Register



- Output of previous FF is connected to the input of the next one via a combinational CKT.
- Binary word $B_0B_1B_2B_3$ is applied through same combinational CKT.
- CKT work on two modes → Shift mode & Load mode.

Load mode

- When Shift/Load line is Low (0), AND gates 2, 4, and 6 become Active. They will pass B_1 , B_2 and B_3 bits to corresponding flip flops.
- On Negative edge of clock, $B_0B_1B_2B_3$ will be loaded into the corresponding flip flops. Thus parallel loading takes place.

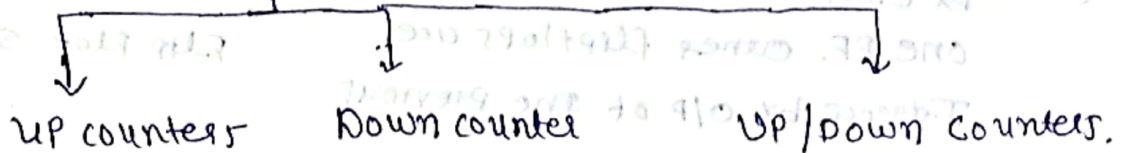
Shift mode: When Shift/Load is high (1), AND gates 2, 4, 6 become inactive so parallel loading becomes impossible.

- AND gates 1, 3 and 5 become active. So on application of clock pulse, shifting of data from left to right occurs.

Thus parallel in serial out operation takes place.

- Q1:- What is Counter? Give Classification of Counter
- Give types of counters.
Give Application of counters.
- Counter is sequential ckt which is used to count pulses.

Counters (Synchronous / Asynchronous)



Asynchronous counters:- (Ripple counter or series counter)

External Clock pulse is applied to only one FF. Output of the preceding Flip Flop is given to the Clock of the next Flip Flop.

→ Counting is slow.

Synchronous counters:- External clock pulse is applied to All Flip Flop simultaneously.

→ counting is Fast.

→ Example:- Ring counter, Johnson counter

Based on counting order.

Up counter:- counts the pulses in Ascending order (उपरोक्त क्रम में)

Down counter:- counts the pulses in Descending order (उपरोक्त क्रम में)

UP/down counter:- counts the pulses in both order.

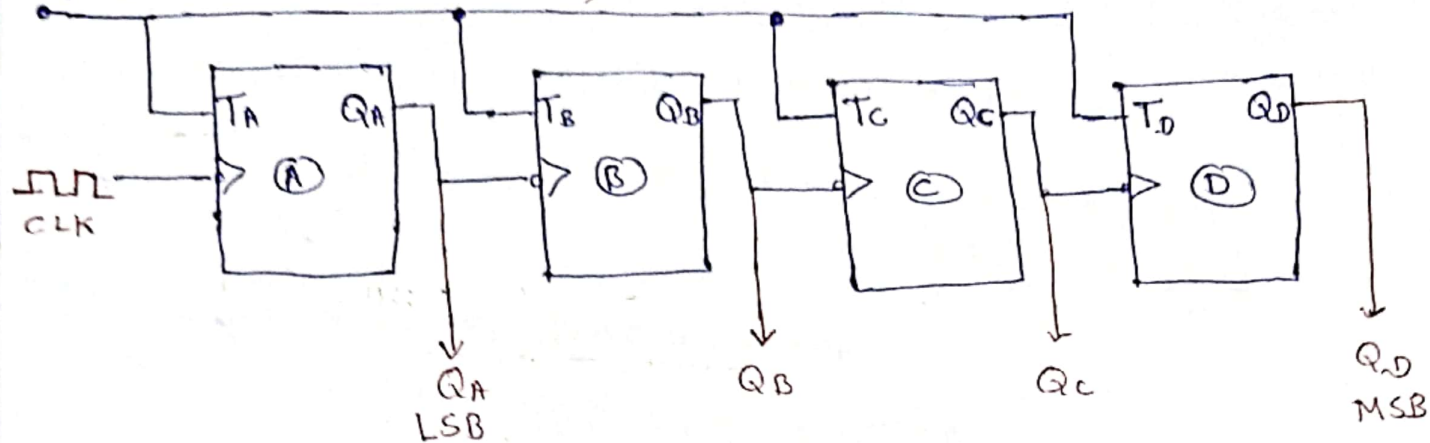
Ascending as well as Descending.

Application

- ① Object counter
- ② Digital Timer
- ③ counting of Packing material
- ④ in Washing machine, Microwave oven.
- ⑤ in Digital clock, frequency meter
- ⑥ Synchronous waveform generator.

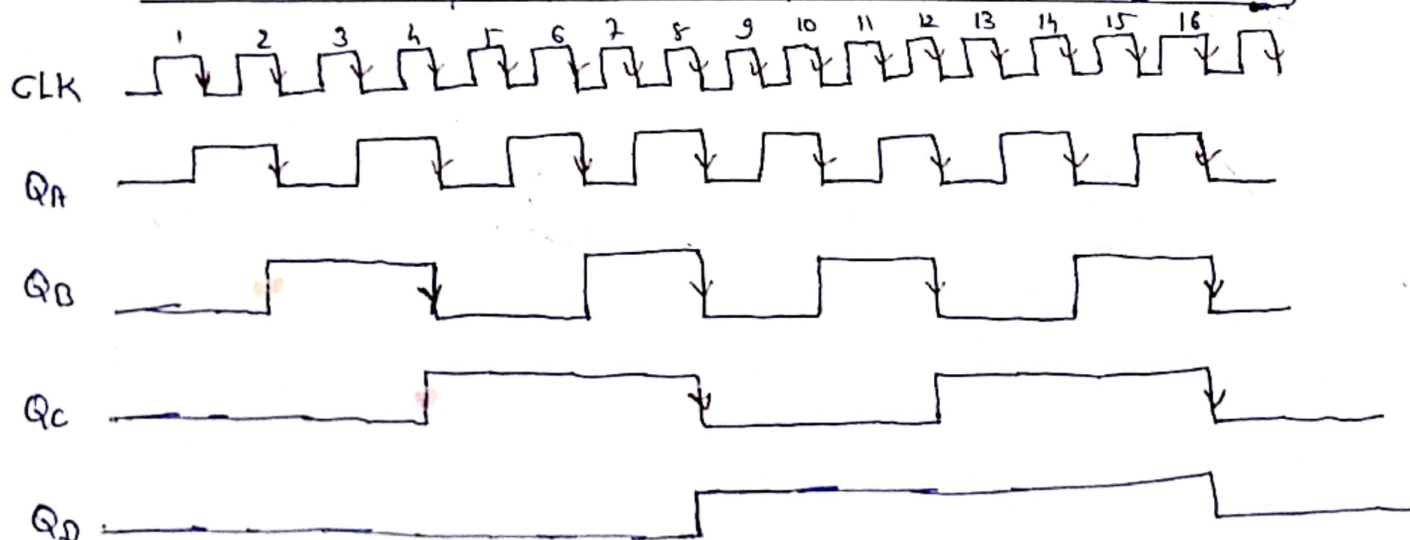
A Bit Asynchronous Counter (UP counter) (Ripple counter)

Logic 1



Truth Table

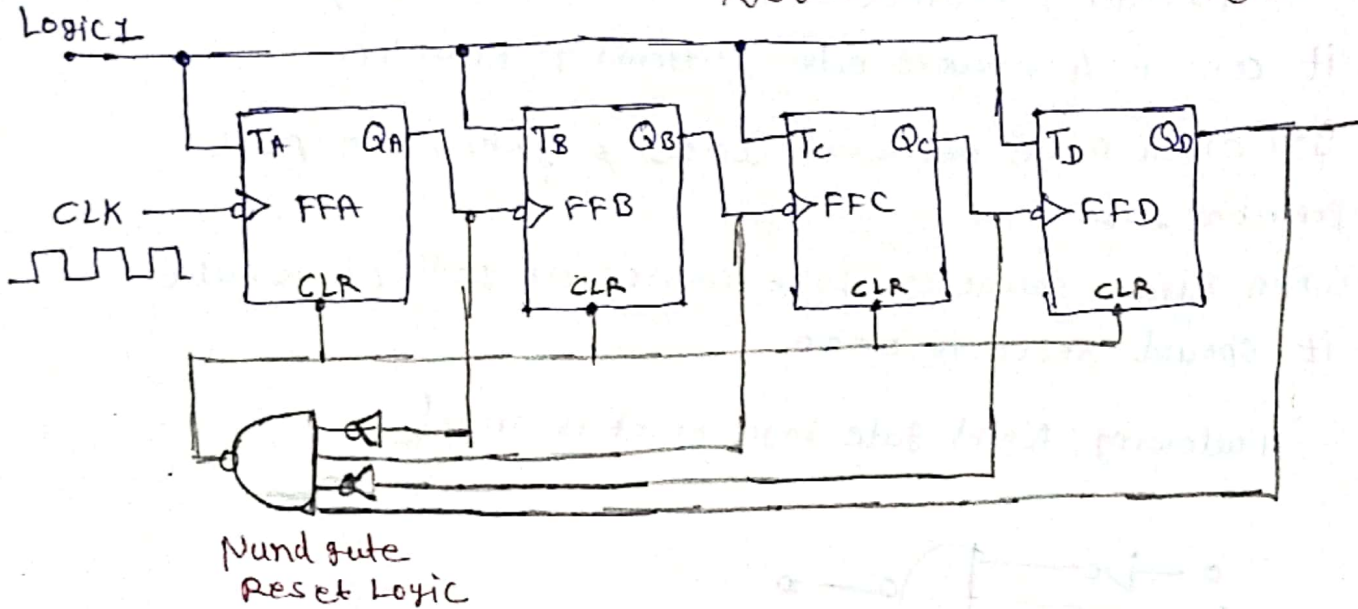
CLK	Q _D	Q _C	Q _B	Q _A	Decimal equivalent	State
initially	0	0	0	0	0	1
1↓	0	0	0	1	1	2
2↓	0	0	1	0	2	3
3↓	0	0	1	1	3	4
4↓	0	1	0	0	4	5
5↓	0	1	0	1	5	6
6↓	0	1	1	0	6	7
7↓	0	1	1	1	7	8
8↓	1	0	0	0	8	9
9↓	1	0	0	1	9	10
10↓	1	0	1	0	10	11
11↓	1	0	1	1	11	12
12↓	1	1	0	0	12	13
13↓	1	1	0	1	13	14
14↓	1	1	1	0	14	15
15↓	1	1	1	1	15	16
16↓	0	0	0	0	0	1



4 Bit Asynchronous counter (UP counter) (Ripple counter)

- 4 bit Ripple UP counter, we need 4 FF.
- it has $2^4 = 16$ states. From 0000 to 1111
(0)₁₀ to (15)₁₀
- Initially all FF have zero O/P.
 $Q_D Q_C Q_B Q_A = 0000$
- All FF are negative edge triggered. CLK is applied to the CLK input of FFA.
- Q output of every FF is applied to the CLK input of Next FF.
- After 1111, the output again become 0000 and operation repeats itself.

Q:- Decode ripple counter OR BCD ripple counter OR MOD 10 counter

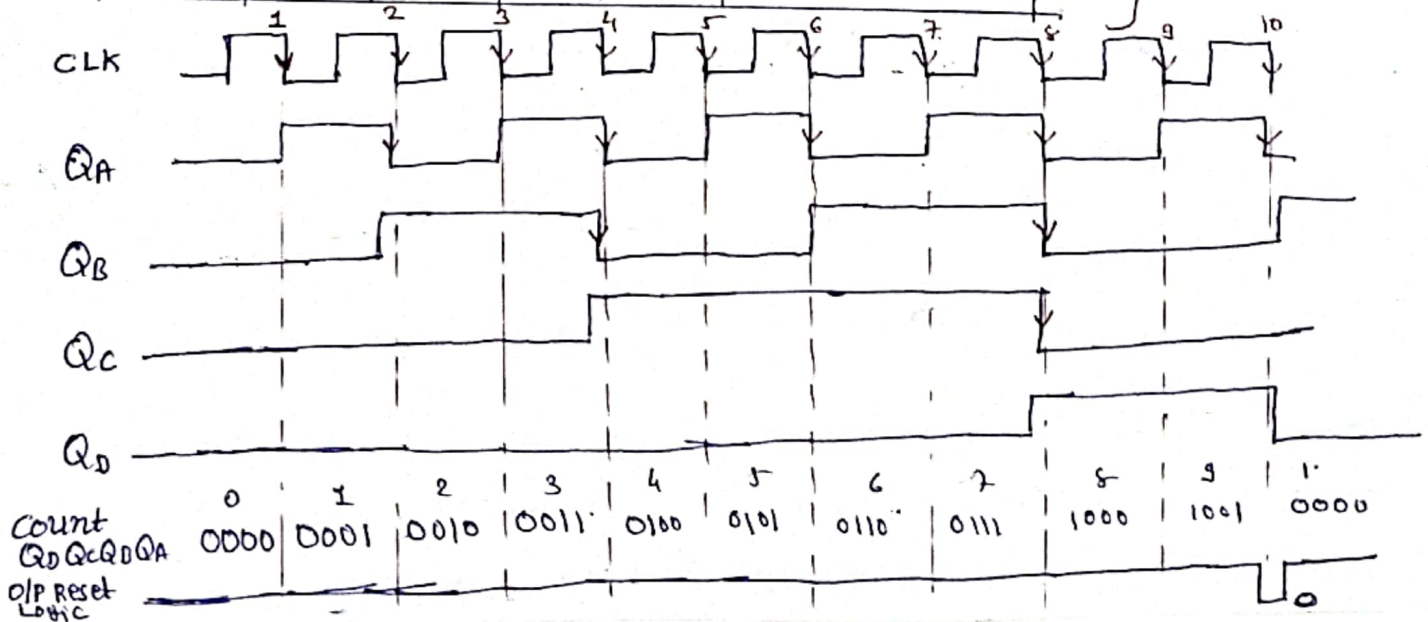


Truth Table

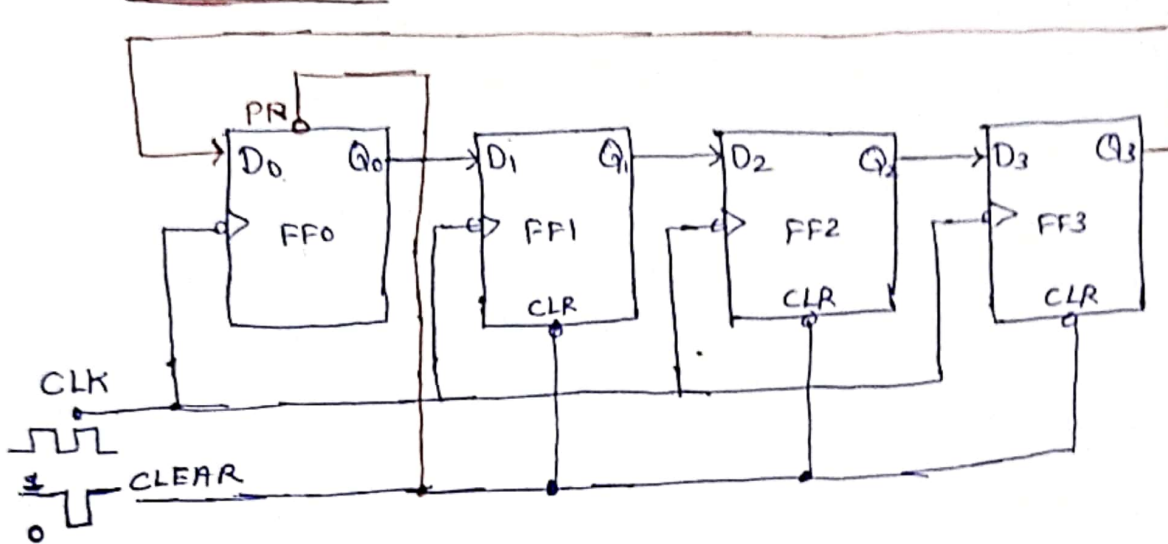
CLK	Q _D	Q _C	Q _B	Q _A	Output of Reset Logic
0	0	0	0	0	1
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
-	1	0	1	0	0
-	1	0	1	1	0
-	1	1	0	0	0
-	1	1	0	1	0
-	1	1	1	0	0
-	1	1	1	1	0

Valid states

Invalid states



Q:- Ring Counter



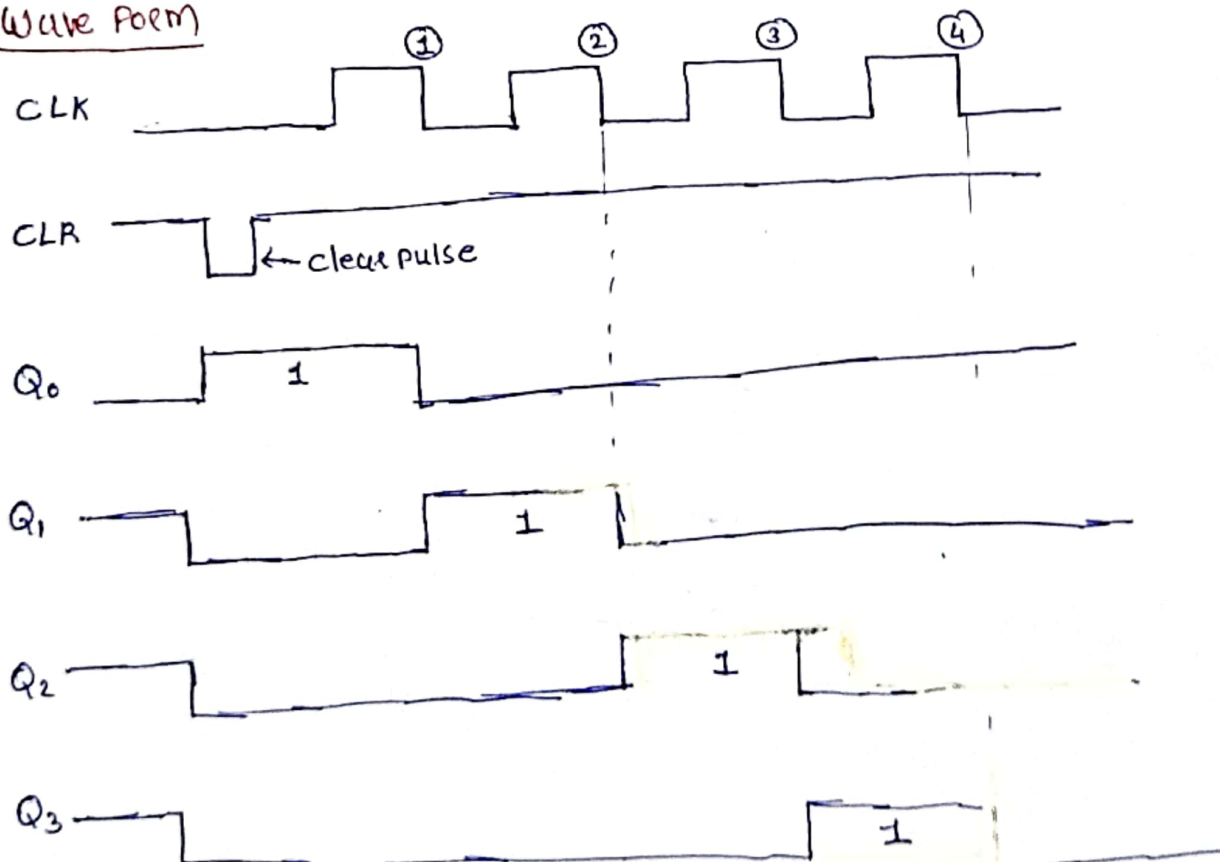
Summary of Operation

CLR	CLK	Q ₀	Q ₁	Q ₂	Q ₃
	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓	0	0	0	1
1	↓	1	0	0	0

← FF0 Preset, others cleared.

Presetted 1 follows a circular path to form a ring.

Wave Form



Ring Counter

- One application of shift Register is Ring Counter.
- O/P of FF3 is connected to data input D_0 of FF0.
- No of states = No of flip flop.

Operation

- Initially low clear (CLR) pulse is applied to all FF.
So FF3, FF2, FF1 will Reset but FF0 will be set to 1

$$\text{So } Q_3 Q_2 Q_1 Q_0 = 0001$$

- Then clear Terminal is made inactive by apply high Level to it.

- On first Negative going CLK edge

→ FF1 will be set because $Q_0 = D_1 = 1$

→ FF0 will reset because $D_0 = Q_3 = 0$ & No change in FF2 & FF3

$$\text{So } Q_3 Q_2 Q_1 Q_0 = 0010$$

After 2nd Negative going CLK edge $Q_3 Q_2 Q_1 Q_0 = 0100$

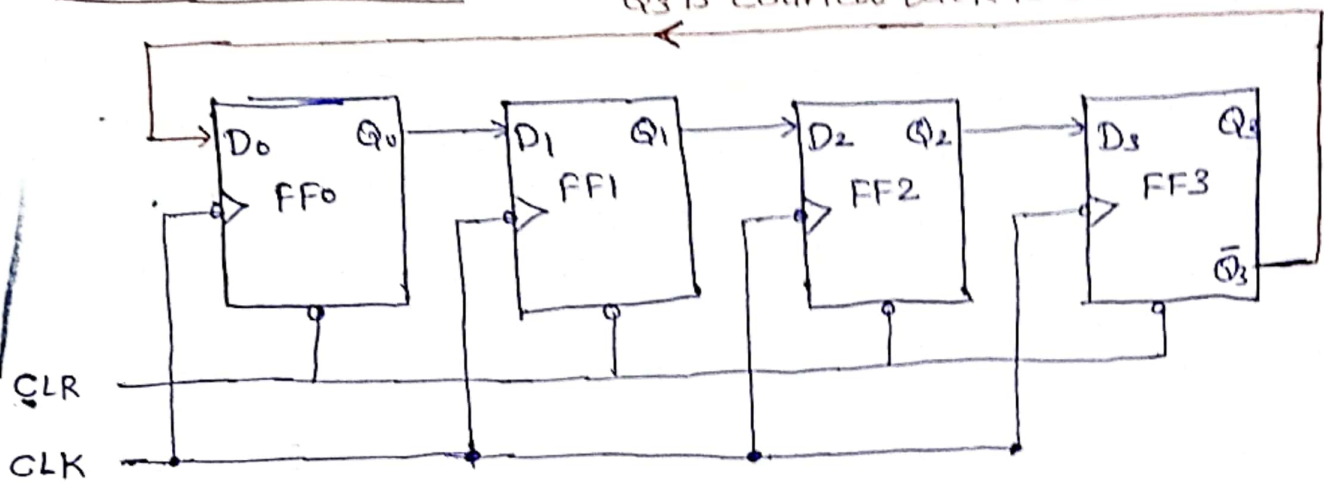
" 3rd " " " " $Q_3 Q_2 Q_1 Q_0 = 1000$

" 4th " " " " $Q_3 Q_2 Q_1 Q_0 = 0001$

Hence operation repeats.

Johnson Counter

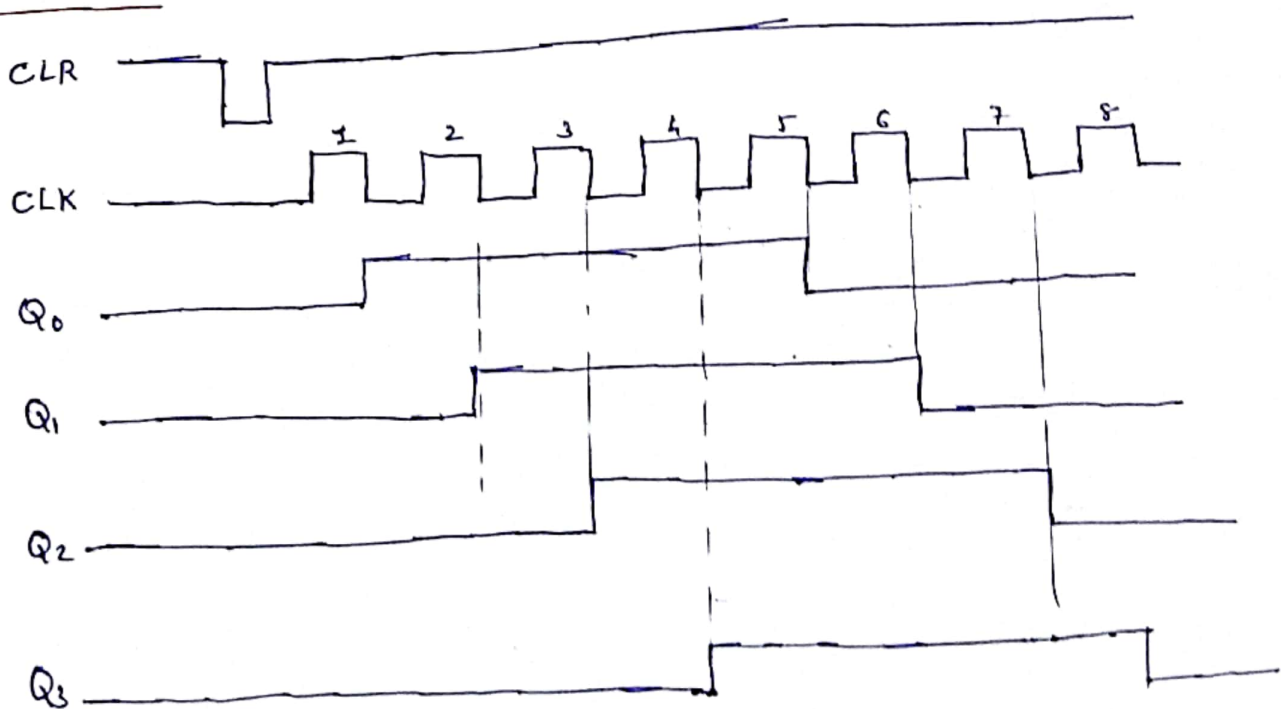
Q_3 is coupled back to D_0



Summary of operation

$\overline{CLR}$	CLK	Q_0	Q_1	Q_2	Q_3	State number
	initially	0	0	0	0	1
1	↓	1	0	0	0	2
1	↓	1	1	0	0	3
1	↓	1	1	1	0	4
1	↓	1	1	1	1	5
1	↓	0	1	1	1	6
1	↓	0	0	1	1	7
1	↓	0	0	0	1	8
1	↓	0	0	0	0	1

Wave Form



Johnson counter

- in Johnson counter output is cross coupled to the input $\bar{Q}_3$ is connected to the Data input D_0 of FFO.
- NO of States = $2 \times \text{NO of Flipflop}$
- All FF are negative edge triggered. Clock pulse are applied to all simultaneously, (Parallel).

Operation :-

Initially $\overline{\text{Clear}}$ is Low, so outputs are

$$Q_0 Q_1 Q_2 Q_3 = 0000$$

↓ On 1st First Negative edge of clock pulse

$$Q_0 Q_1 Q_2 Q_3 = 1000$$

↓ on second negative edge of clock pulse

$$Q_0 Q_1 Q_2 Q_3 = 1100$$

↓ On Third negative edge of clock pulse.

$$Q_0 Q_1 Q_2 Q_3 = 1110$$

↓ after 4th negative edge of clock pulse

$$Q_0 Q_1 Q_2 Q_3 = 1111$$

↓ after 5th negative edge of clock pulse

$$Q_0 Q_1 Q_2 Q_3 = 0111$$

This operation will continue till

$$Q_0 Q_1 Q_2 Q_3 = 0000$$

Total 8 output states.

